

Vallepu Anil Sahith

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SKILLS SUMMARY

- **Programming Languages:** C++, Python, JavaScript, Typescript, SQL, Matlab.
- **Developer Tools:** VS Code Studio, PyCharm, Spyder, Cursor, Figma, Postman, Github.
- **Web Technologies:** React, Next.js, Flask, FastAPI, WebSocket, Streamlit, Gradio, Beautiful Soup.
- **Databases and Cloud:** MySQL, PostgreSQL, MongoDB, Redis, FAISS, Pinecone, Git, AWS, GCP, Docker.
- **AI/ML Technologies:** Scikit-learn, Keras, TensorFlow, PyTorch, OpenCV, NLP, RAG, MCP Servers.
- **Familiar with:** Data Structures and Algorithms, System Design, Embedded Systems, IoT and Edge Computing, LLM Fine-Tuning and Inference Optimization, Quantum Computing.

WORK EXPERIENCE

- **Noccarc Robotics Pvt Ltd.** Jun 2024 - Apr 2025
AI Engineer Intern Pune, Maharashtra
 - Developed a real-time arrhythmia detection system for embedded patient monitors using a combined multichannel hybrid model (ResNet+CBAM+GRU with separable convolutions and attention layer) to generate predictive masks for 18+ arrhythmia classes, achieving over 95 percent accuracy on publicly available PhysioNet database.
 - Architecture includes data flow from ECG sensors, front-end display, back-end processing, and model inference pipelines for seamless embedded hardware operation. Model inference occurs in real time every 5 seconds, generating alarms if any abnormality is detected, all while consuming less than 10 percent of single-core CPU resources.
 - System enables real-time detection of heart abnormalities, generating nearly 20+ distinct alarms via the model, facilitating immediate clinical decision-making and reducing response times in critical care settings.
- **Carelon Global Solutions, (Elevance Healthcare)** Jun 2023 - Aug 2023
Summer Intern - AI/ML Bangalore, Karnataka
 - Integrated PyTesseract for Optical Character Recognition (OCR) to extract text, OpenCV for image pre-processing to improve OCR accuracy, and YOLOv8 for object detection identifying and isolating specific regions of interest, processing nearly 700 documents per hour with 95 percent accuracy.
 - Implemented multi-threading and multi-processing techniques, Delivered a comprehensive solution that integrates seamlessly with existing data workflows, providing a reliable and scalable solution to automate data extraction and reduce processing time and cost by 90 percent.

PROJECTS

- **MiniAI-As-A-Service App:** Built a comprehensive full-stack application that transforms CSV data analysis into an intelligent, automated experience. Enabled users to upload CSV files, generate instant statistical insights, train ML models, and get AI-generated summary via a professional web interface for data scientists and analysts. **Tech stack:** Next.js, Typescript, Tailwind CSS, Python, Scikit-learn, XGBoost, FastAPI, PostgreSQL, Redis, Docker, OpenRouter.
- **SlackBot Content Creation AI Assistant:** The SlackBot Content Creation Assistant is an AI-powered workflow automation tool designed to streamline the content research and planning process for content teams, SEO specialists, and digital marketers. By integrating directly into Slack, the bot eliminates hours of manual keyword research, competitive analysis, and content planning, to minutes. **Tech stack:** Python, Supabase(PostgreSQL), Upstash(Redis), Brave Search API, SerpAPI, Scikit-learn, OpenAI Embeddings, Groq, SendGrid, Docker, Render.
- **HSN Code Classification System:** Developed an intelligent HSN (Harmonized System Nomenclature) Code Classification System using RAG and Knowledge Graphs. The chatbot system will help businesses accurately identify the correct 8-digit HSN codes for products, addressing the challenges of complex hierarchies, similar product descriptions, and lack of domain expertise, **Tech stack:** Python, Streamlit, PyPDF2, NetworkX, Faiss, Sentence Transformers, OpenRouter.
- **Dynamic AI Exam-Prep App:** Developed an AI-powered adaptive study calendar application using a custom dynamic AI agent orchestration system to personalize the preparation for the NCLEX exam. Implemented real-time student performance analytics, intelligent scheduling of calendars, and contextual AI chatbot tutoring to maximize learning outcomes. **Tech stack:** Next.js, Typescript, Tailwind CSS, Chart.js, MongoDB, Google Generative AI, OpenRouter.

POSITION OF RESPONSIBILITY

- **Contributed as a member of Big Data, Analytics and Consulting Cell, BDACC** Nit Warangal
Led the organization of multiple events,including summer and winter schools for Data Science.Aug 2022 - May 2023
- **Captained the Inter-NIT Volleyball team** Nit Warangal
Organized more than 10 different volleyball tournaments for students within the college. Aug 2022 - May 2023

ACHIEVEMENTS

- Secured JEE Mains (India) 99.3rd percentile - was among 0.7 percent of all who appeared for this competitive exam.
- Ranked within the top 100 projects at the NASA AMES Space Settlement Contest in San Juan, Puerto Rico (2017)
- Solved 300+(Medium to Hard) DSA problems on Leetcode.